



Pocket Powerbox **Advance**

Serial Command Language

Firmware >=v 2.9.x (reviewed Oct 2023)

Abbreviations used:

nnnn.. = one or more digits

b = Boolean (0 or 1 digit)

Connection Settings: 9600, 8N1
(All commands should be terminated by new line /n)

Command	Description	Response
P#	Device Status	PPBA_OK
PE:bb	Set Power Status on boot. 1st number represents the quad output. 2 nd number represents the adjustable out output. (0=OFF, 1=ON).	PE:1
PE:99	Report Power Status on boot. Ignore the last 2 numbers.	PE:nnnn
P1:b	ON/OFF Power 4x12V Outputs, (0=OFF, 1=ON)	P1:b
P2:n	ON/OFF Adjustable Output, (0=OFF, 1=ON) n can also accept values of: 3, 5, 7, 8, 9, 12 to adjust output to specific voltage (stored in EEPROM and retrieved during boot)	P2:n
P3:nnn	PWM Duty Cycle Power % (DewA) [n=0-255] (0-100%)	P3:nnn
P4:nnn	PWM Duty Cycle Power % (DewB) [n=0-255] (0-100%)	P4:nnn
PF	Reboot Device, Reload Firmware	[none]
PA	Print Power and Sensor Readings	[Check table below]
PS	Prints Power Consumption Statistics	PS : averageAmps : ampHours : wattHours : uptime_in_milliseconds
PC	Print Power Metrics *Current is represented in Amps and does not require conversion	PC:total_current : current_12V_outputs : current_dewA : current_dewB : uptime_in_milliseconds
PR	Prints discovered I2C devices plugged to EXT port	PR:HDC:DHT:XS if there is a discovered device command will output its name HDC = temp/humidity sensor TI HDC1050 DHT =stock temp/humidity sensor AM2301 XS: eXternal Motor (stepper) Controller
DA	Print Auto Dew Aggressiveness. from 0 to 255 (210 default value). A higher value sets higher power to dew heaters when humidity is at x level.	DA:nnn
PD:b	Enable/Disable Auto Dew Feature (0=OFF, 1 =ON)	PD:b
PD:nnn	Set Auto Dew Aggressiveness value [1-254]	PD:nnn

DS:b	If set to 1: Store DewA,B ports current values into EEPROM and set a retrieval value, This value is considered during device boot. If set to 0: disable value retrieval on device boot.	DS:b
DS:99	Read dew ports retrieve flag on device boot. 1= YES, 0 = NO.	DS:1 or DS:0
PV	Firmware Version	n.n.n
PU:b	Switch ON /OFF USB2 Hub (Controls dual USB2 ports <u>only</u> on PPBADV Gen2)	PU:b
PI	Reset I2C channel	PI:1
CH:nn	Calibrate Humidity Sensor. Set offset value (positive or negative)	CH:nn
CT:nn	Calibrate Temperature Sensor. Set offset value (positive or negative)	CT:nn
CR	Reads Calibration Offsets	CR:temp_offset:humidity_offset
PL:b	ON/OFF Led Indicator (0=OFF, 1=ON) value is not stored in memory	PL:b
XS	eXternal Motor Controller commands	check table below

Transmit: PA

Receive: PPBA:12.2:0.5.22.2:45:17.2:1:1:120:130:1:0:1

Meaning:

PPBA:voltage:current_of_12V_outputs_:temp:humidity:dewpoint:quadport_status:adj_out
put_status:dewA_power:dewB_power:autodew_bool:pwr_warn:pwradj

PPBA	Device Name
voltage	Input Voltage in Volts (decimal) e.g 12.2
current	Quad 12V output sens current 0-1024 (need to convert to Amps by dividing by 65 *for compatibility issues with PPB. Better use output from PS command
temp	Temp in Celsius Degrees (decimal) e.g 23.2
humidity	Relative Humidity in % (integer) e.g 59
dewpoint	Dewpoint in Celsius Degrees (decimal) e.g 14.7
quadport_status	Boolean 0 or 1 (1 means port is ON, 0 means port is OFF)
adj_output_status	Adjustable Output. Boolean 0 or 1 (1 means port is ON, 0 means port is OFF)
dewA_power	Power of DewA channel - duty cycle 0-255
dewB_power	Power of DewB channel - duty cycle 0-255
autodew_bool	Boolean for autodew function (controls power of both Dew channels): 0 is OFF, 1 is ON
pwr_warn	Boolean. 1 means power alert (short wire detection / output overload). This is a generic flag for any 12V and DewA, DewB power outputs.
pwradj	Adjustable Output: Selected voltage in EEPROM (3,5,7,8,9,12)

eXternal Motor Controller (XMC) Command Set (through powerbox)

Below command language should be given from the USB serial (9600/8N1) of the Powerbox.
The I2C command language is not described here.

XS: [command number] #value

Command	Description	Response
XS	Status	200 response = discovered 000 response = non present E.g XS:200:0
XS:1	Motor is running (1) or is idle (0)	XS:1#b
XS:2	Motor current position (long value)	XS:2#nn...
XS:3#nn..	Move to position (long value)	XS:3#nn
XS:4#nn..	Goto +steps from current position (long value)	XS:4#nn..
XS:5#nn..	Set a new motor position (long value)	XS:5#nn
XS:6	Halt motor	XS:6#1
XS:7#nn..	Set / Get motor max speed	XS:7#nn..
XS:8#b	Set motor reverse (0 = normal, 1= reverse)	XS:8#b
XS:9#n	Set Microstepping drive 1 = FULL 2 = HALF 3 = 1/4 4 = 1/8	XS:9#n
XS:10#nn..	Set Motor Backlash (enables backlash compensation). Set 0 disabled the backlash compensation	XS:10#nn